The empirical recurrence for OEIS A296325: recovering a conjecture that is not written down

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Abstract

OEIS A296325 records "Empirical recurrence of order 83", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 270$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296325 is "Number of nX6 0..1 arrays with each 1 horizontally, vertically or antidiagonally adjacent to 2 neighboring 1s.". It has offset 1 and begins

$$1, 26, 122, 562, 3569, 20088, 112235, 643541, \ \ldots$$

The entry states:

> Empirical recurrence of order 83 (see link above)

As of the "Last modified" line on the live entry (Dec 10 2017 11:46:31, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 270$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 270$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 540$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{83} c_i\, a(n-i),$$

c_1	11	c_{22}	−19762083	c_{43}	178737999	c_{64}	1160261
c_2	−55	c_{23}	31475075	c_{44}	63580329	c_{65}	1867318
c_3	241	c_{24}	−25257059	c_{45}	−170582996	c_{66}	−1207092
c_4	−908	c_{25}	−4808878	c_{46}	34833459	c_{67}	−554710
c_5	3067	c_{26}	40072248	c_{47}	105268561	c_{68}	741993
c_6	−9711	c_{27}	−46095292	c_{48}	−49337668	c_{69}	−25955
c_7	27813	c_{28}	6624672	c_{49}	−72283002	c_{70}	−274332
c_8	−74909	c_{29}	51643044	c_{50}	55473970	c_{71}	105926
c_9	183330	c_{30}	−87412493	c_{51}	49371449	c_{72}	51753
c_{10}	−420431	c_{31}	70010667	c_{52}	−65882316	c_{73}	−49597
c_{11}	885669	c_{32}	12029366	c_{53}	−13000515	c_{74}	2907
c_{12}	−1729628	c_{33}	−124523001	c_{54}	53812617	c_{75}	11319
c_{13}	3048325	c_{34}	159743876	c_{55}	−14362907	c_{76}	−4525
c_{14}	−4736989	c_{35}	−24813174	c_{56}	−26874690	c_{77}	−697
c_{15}	6154528	c_{36}	−190439398	c_{57}	18655128	c_{78}	1036
c_{16}	−6039876	c_{37}	247632502	c_{58}	7656302	c_{79}	−252
c_{17}	3239849	c_{38}	−39288059	c_{59}	−11762771	c_{80}	−46
c_{18}	1921863	c_{39}	−228985052	c_{60}	−952079	c_{81}	41
c_{19}	−6239421	c_{40}	254106518	c_{61}	6291602	c_{82}	−10
c_{20}	5219130	c_{41}	−3606971	c_{62}	−545853	c_{83}	1
c_{21}	4306128	c_{42}	−230313050	c_{63}	−3662481		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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